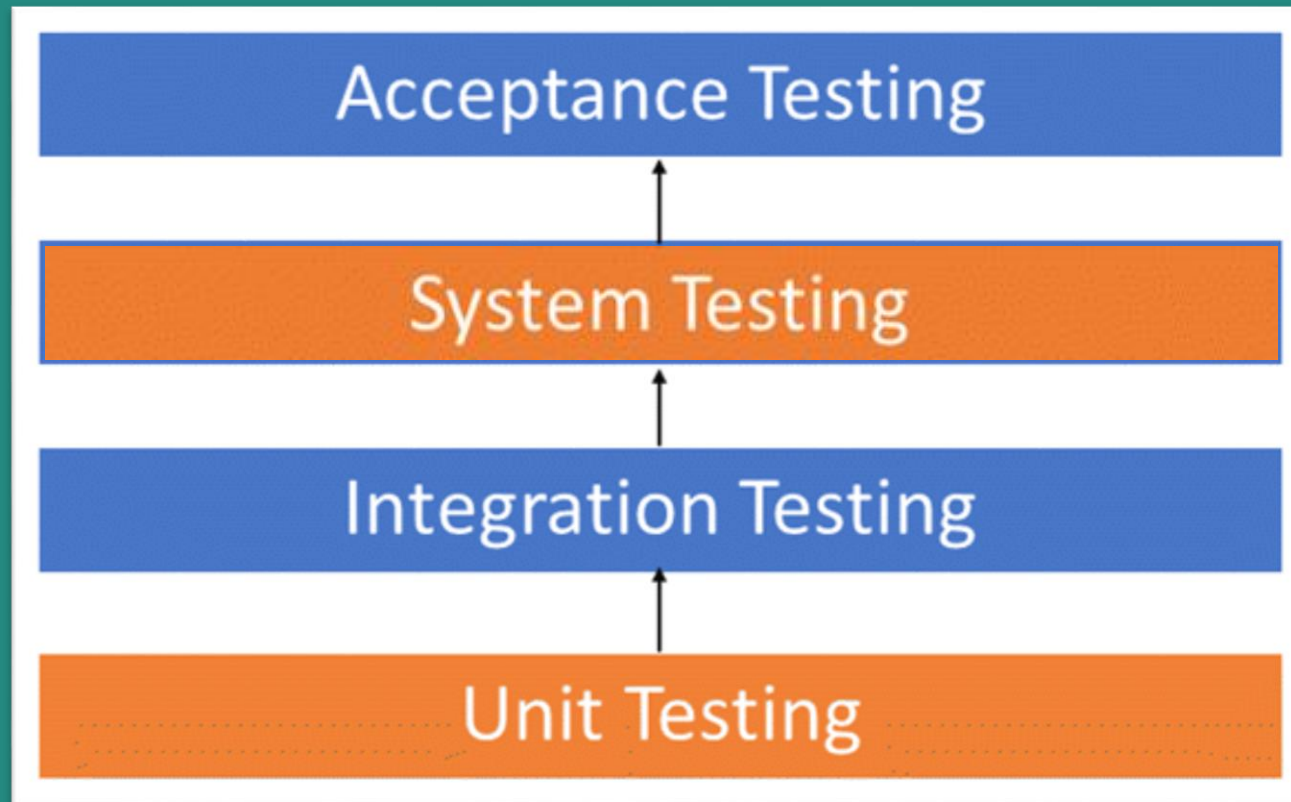


Software Testing

Types of Testing

- *ISTQB / ISEB Foundation Exam Practice*
- *<https://www.guru99.com/software-testing.html>*

Contents



Unit Testing

- individual **units** of a software are tested
 - a line of code, a method, or a class
- to validate that each unit performs as **expected**
- done by the **developers** who writes the code.

UT Example (1)

```
Prints (int a, int b) {  
    int result = a+ b;  
    If (result> 0)  
        Print ("Positive", result)  
    Else  
        Print ("Negative", result)  
}
```

----- Printsum is a function

----- End of the source code

```
1 ▾ Prints (int a, int b) {  
2   int result = a+ b;  
3   If (result> 0)  
4       Print ("Positive", result)  
5   Else  
6       Print ("Negative", result)  
7   }  
8
```

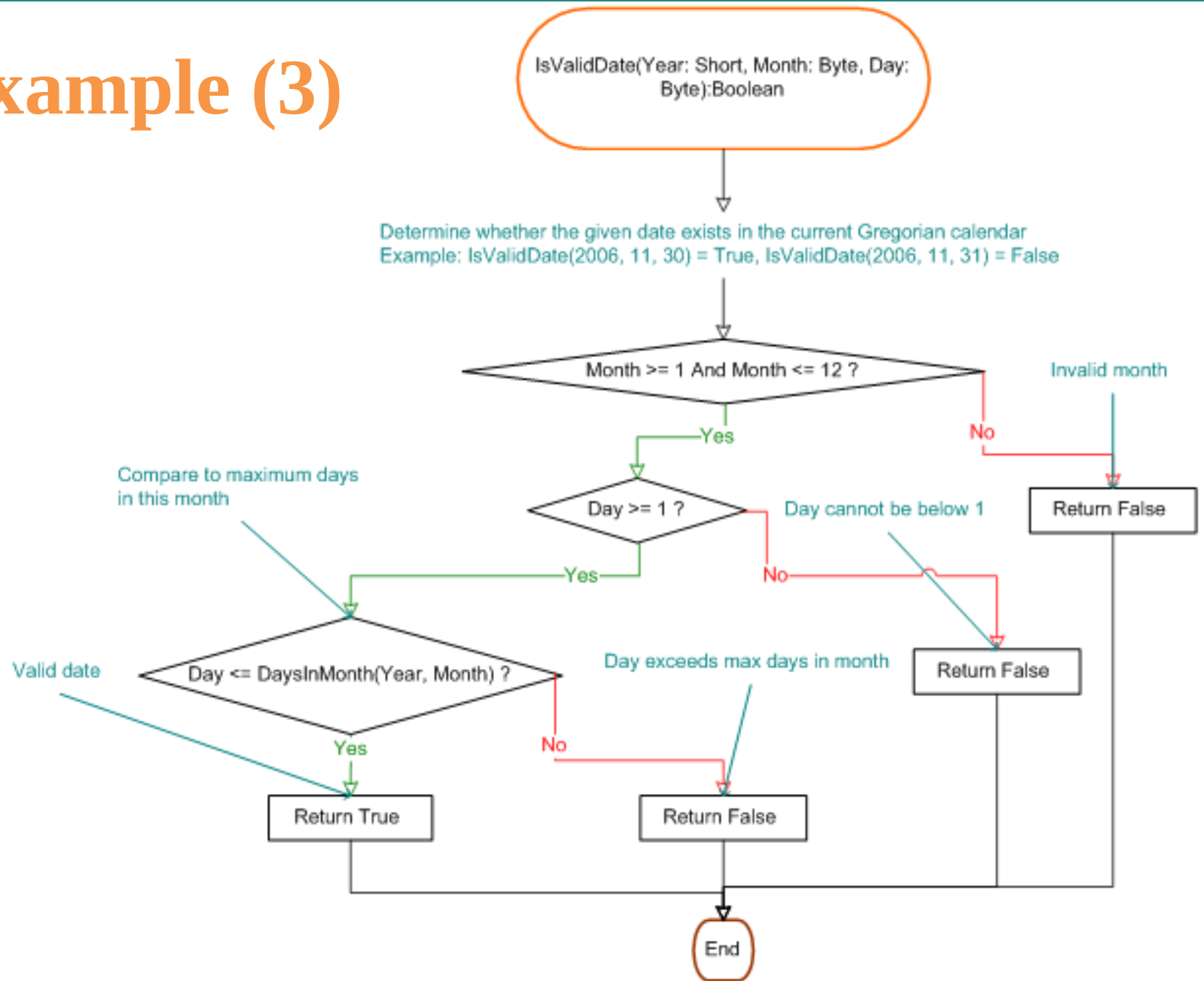
```
1 ▾ Prints (int a, int b) {  
2   int result = a+ b;  
3   If (result> 0)  
4       Print ("Positive", result)  
5   Else  
6       Print ("Negative", result)  
7   }  
8
```

a=? b=?

UT Example (2)

```
Demo(int a) {      1.  
    If (a> 5)      2.  
        a=a*3      3.  
    Print (a)      4.  
}                  5.
```

UT Example (3)



Integration Testing

- what the set can perform that is not possible individually
- more than one (tested) component
- communication between components
- done by designers, analysts, or independent testers

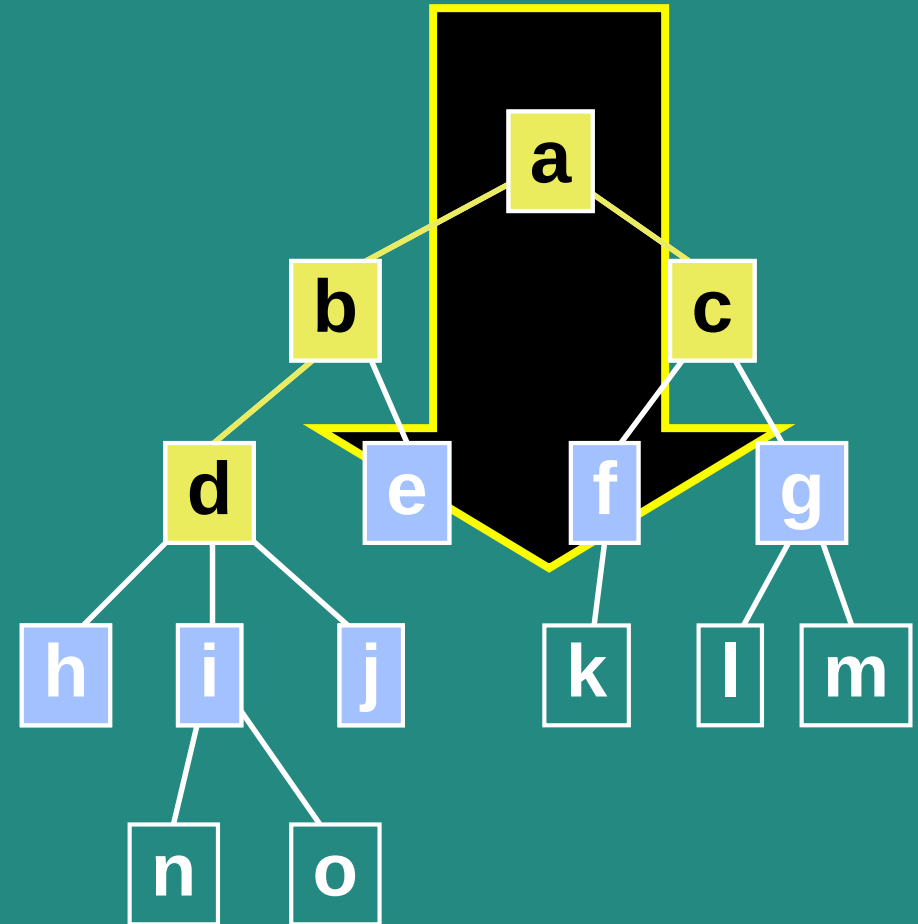


Big-Bang Integration

- All the components or modules are integrated together at once and then tested as a unit
- If all of the components in the unit are not completed, the integration process will not execute.

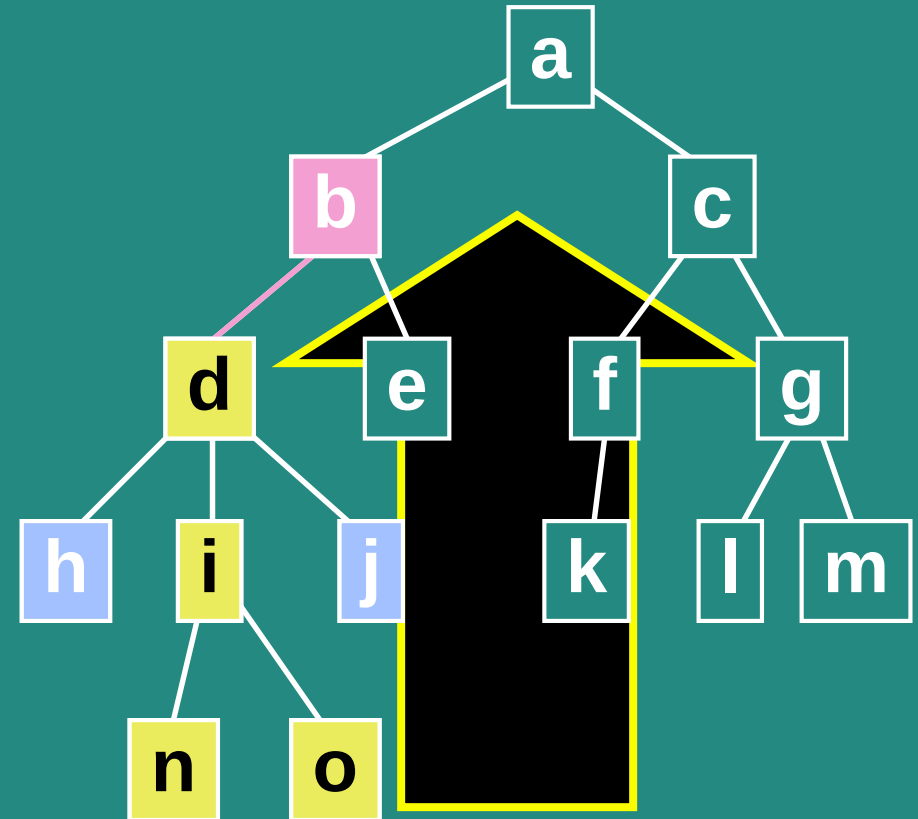
Top-Down Integration

- Baselines:
 - baseline 0: component a
 - baseline 1: a + b
 - baseline 2: a + b + c
 - baseline 3: a + b + c + d
 - etc.
- Need to call to lower level components not yet integrated
- Stubs: simulate missing components

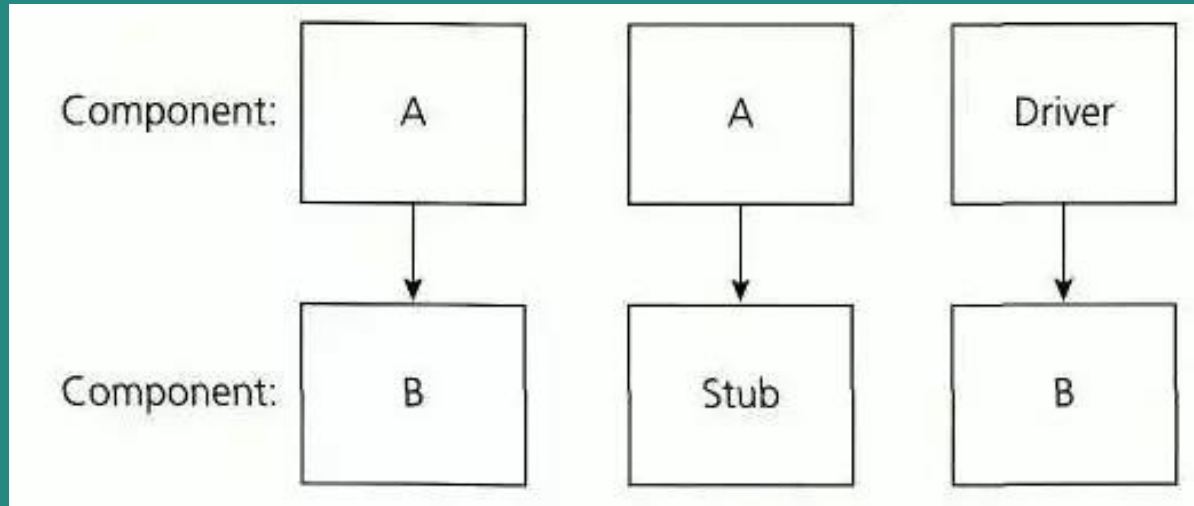


Bottom-up Integration

- Baselines:
 - baseline 0: component n
 - baseline 1: $n + i$
 - baseline 2: $n + i + o$
 - baseline 3: $n + i + o + d$
 - etc.
- Needs drivers to call the baseline configuration
- Also needs stubs for some baselines



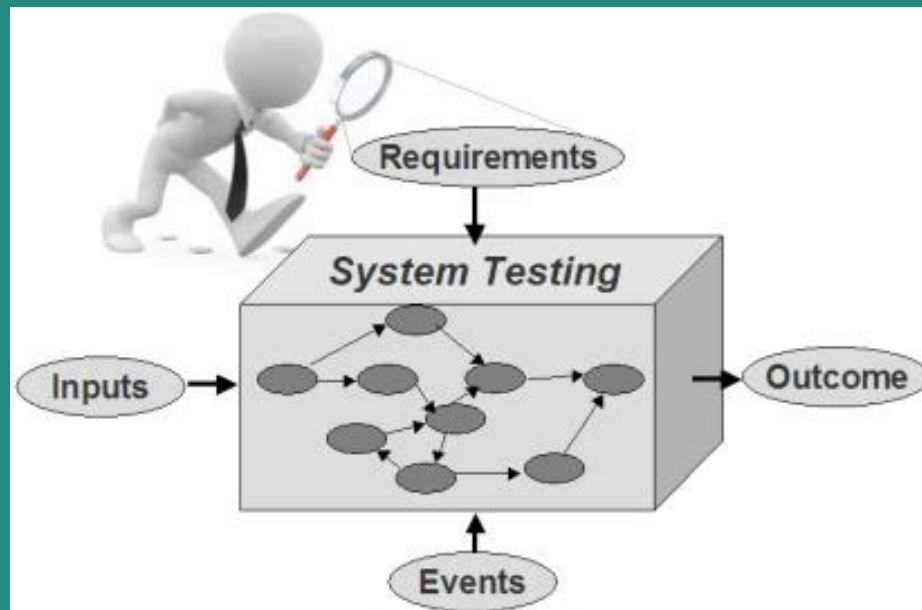
Stubs and Drivers



- **Stubs and Drivers:** substitutes for the missing models.
- **Stub:** Is called by the Module under Test.
- **Driver:** Calls the Module to be tested.

System testing

- functional
- non-functional
- done by independent test group



Non-functional system testing

- performance: speed, response time,
- security
- stability
- reliability
- scalability
- ...

Performance Test Cases

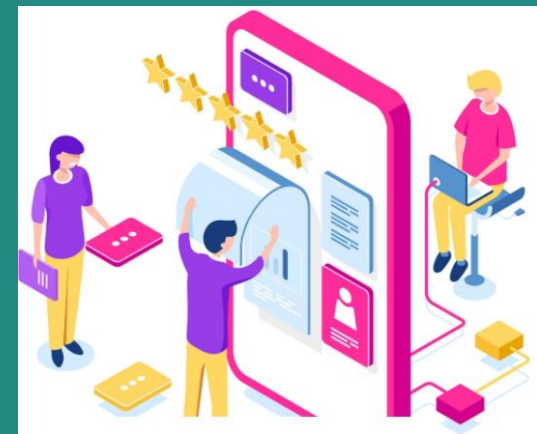
1. Verify **response time** is not more than 4 secs when 1000 users access the website simultaneously.
2. Verify **response time** of the Application Under Load is within an acceptable range when the network connectivity is slow
3. Check the **maximum number of users** that the application can handle before it crashes.
4. Check **database execution time** when 500 records are read/written simultaneously.
5. Check **CPU and memory usage** of the application and the database server under peak load conditions

Software Requirements Specification

1	INTRODUCTION	5
	1.1 Purpose.....	5
	1.2 Scope.....	5
	1.3 Definitions, acronyms and abbreviations	5
	1.4 References	6
	1.5 Overview	6
2	FUNCTIONALITY	7
	2.1 User Interface.....	7
	2.2 "Close" function	7
	2.3 "Clear" function	8
	2.4 Check Date Time function.....	8
3	REABILITY.....	12
4	PERFOMANCE	13
5	TECHNOLOGY RULES AND LIMITATIONS	14
	5.1 Running Environment.....	14
	5.2 Development Environment.....	14
	5.3 Hardware	14

User Acceptance Testing

- Final stage of validation
- Acceptance testing conducted to enable the customer to determine whether or not to accept the system
- User Acceptance Testing is performed by
 - the end user
 - or the customer



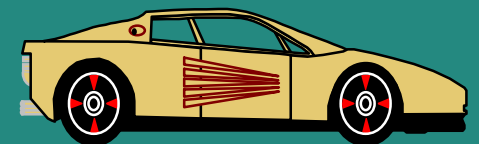
User Acceptance Testing

Acceptance testing
distributed over
this line

20% of function
by 80% of code

80% of function
by 20% of code

System testing
distributed over
this line



Alpha and Beta Testing



- **Alpha/Beta Testing** is a type of acceptance testing

Alpha and Beta Testing: differences

- The main focus of alpha testing is to **simulate** real users
- Alpha Testing is performed by the Testers within the organization
- Alpha Testing is performed at Developer's site
- Beta Testing is performed by the end users (real users, **real** environment)
- Beta Testing is the final test before shipping a product to the customers

Bài tập thực hành

1. Cho ví dụ về Stubs và Drivers trong kiểm thử phần mềm
2. Mô tả Alpha tests và Beta tests mà bạn đã từng trải nghiệm
3. Hiện thực hàm IsValidDate theo sơ đồ giải thuật slides 6 bằng ngôn ngữ C/C++/C#/Java. Liệt kê các trường hợp để kiểm thử hàm này sao cho tất cả các dòng code đều được thực thi ít nhất 1 lần

Bài 01

- Viết ra yêu cầu hợp lệ của field DD, MM và YYYY
- Viết ra yêu cầu chức năng của button Calculate

The image shows a Windows application window titled "Calculate next DAY !". The window contains two main sections: "Current DAY" and "Next DAY". Each section has three input fields for DD, MM, and YYYY. A "Calculate !" button is positioned between the two sections.

Current DAY

DD	MM	YYYY

Calculate !

Next DAY

DD	MM	YYYY

Bài 02

- Dựa vào Bài 01 để thiết kế các trường hợp kiểm thử (test case) theo mẫu

STT	Mô tả	Dữ liệu kiểm thử	Kết quả mong đợi	Kết quả thực thi chương trình
1.				
2.				
3.				
4.				